Module 16: Capstone Systems Synthesis - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Connect concepts from at least three modules in one explanation.
2. Trace a mechanism across biological scales.
3. Identify feedback loops, constraints, and tradeoffs.
4. Match evidence types to claims at different scales.
5. Revise a systems explanation using new evidence.

Topics

- Biology across scales
- Mechanism chains
- Feedback and constraint
- Evidence from multiple modules
- Capstone explanation and revision

Key Terms to Know

- **System** - Interacting parts whose relationships shape outcomes.
- **Scale** - Level of biological organization.
- **Mechanism** - Causal process that explains how something happens.
- **Feedback loop** - Circular causal relationship that amplifies or reduces change.
- **Constraint** - Limit that shapes what a system can do.
- **Tradeoff** - Benefit in one function paired with cost elsewhere.
- **Synthesis** - Combining ideas into a coherent explanation.
- **Evidence chain** - Linked evidence supporting each step of a claim.

Core Contents

1. Connect molecules, cells, organisms, populations, and ecosystems through specific mechanisms, such as a plant variant affecting water use.
2. Explain each causal link. Plant reproduction can alter allele frequencies; plant abundance can alter herbivore food. These require different evidence.
3. Balancing feedback counteracts change; reinforcing feedback amplifies it. Constraints limit outcomes; tradeoffs pair a benefit with a cost.
4. Match evidence to scale: sequences test variants, measurements test water use, seed counts test reproduction, and repeated counts test abundance.
5. Connect at least three course modules. State an alternative explanation and a distinguishing observation; revise the weakest link when evidence conflicts.

Study Tips

1. Complete Lab 17 as a prediction, evidence check, and revision exercise. No internet or special equipment is needed.
2. Answer the learning questions as homework practice; show reasoning and units when a question supplies numbers.
3. Use Practice Test 03, then check its self-contained key and explain any correction in your own words.
4. Connect each answer to a mechanism and evidence, rather than a memorized definition alone.

Connected Lab

- `lab-17_capstone-systems-synthesis.md`

Generated Visuals

- Module 16: Capstone Systems Synthesis Evidence Map
- Module 16: Capstone Systems Synthesis Reasoning Sequence
- Module 16: Capstone Systems Synthesis Retrieval and Lab Check